



How well do protected area networks support taxonomic and functional diversity in non-target taxa? The case of Iberian freshwaters



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ABSTRACT

Protected area networks represent one of the mainstays of global conservation policies and are therefore central to current efforts to maintain biodiversity. However, a major limitation of most conservation strategies is their bias towards particular taxonomic groups and ecosystems, meaning that many taxa and habitats are often only incidentally protected as a by-product of inclusion within reserves. Here we investigate how effectively protected area networks, not specifically designated for freshwaters, support aquatic biodiversity in the Iberian Peninsula (Spain and Portugal), using data for water beetles, surrogates of overall macroinvertebrate diversity in these habitats. We explore the behaviour of different measures (α , β and γ) of both taxonomic and functional diversity at different spatial scales. Overall our findings highlight the contrasting performance of reserve systems in the maintenance of either taxonomic or functional diversity, as well as the importance of spatial scale. Iberian reserves perform relatively well in supporting taxonomic diversity of water beetles at the peninsular scale, but the same protected areas poorly represent functional diversity. Such a mismatch cautions against the use of any one diversity component as a surrogate for others, and emphasizes the importance of adopting an integrative approach to biodiversity conservation in aquatic ecosystems. Furthermore, our results often show contrasting patterns at smaller spatial scales, highlighting the need to consider the influence of scale when evaluating the effectiveness of protected area networks.

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1. Introduction

Protected area networks represent one of the mainstays of worldwide conservation policies and are therefore central to current efforts to maintain biodiversity (Chape et al., 2005). Numerous species are highly dependent on protected areas for their continued persistence; occurring either entirely or largely within their bounds (Jackson and Gaston, 2008). However, the implementation of nature reserves is only the start of the task and evaluating how effective they are is a global research priority to better understand their effectiveness in protecting wider biodiversity (Bertzky et al., 2012). A major limitation of most conservation strategies is their bias towards particular taxonomic groups (Martín-López et al., 2009), meaning that many taxa and habitats are often only incidentally protected as a by-product of their

inclusion within reserves. Freshwaters, for example, are key hotspots of biodiversity (Strayer and Dudgeon, 2010), and are recognized as amongst the most endangered habitats in the world with important needs of protection, research and public awareness (e.g. Dudgeon et al., 2006; Geist, 2011; Kingsford and Neville, 2005; Monroe et al., 2009; Strayer and Dudgeon, 2010). Despite this, very few protected areas have been planned specifically for freshwaters (Abell et al., 2007) and the effectiveness of incidental protection in representing aquatic features and taxa remains poorly and incompletely known (e.g. Abellán et al., 2007). Additionally, the groups which have been the focus of most conservation efforts (e.g. vertebrates or plants) are likely to be poor surrogates for diversity patterns in many freshwater organisms (see Darwall et al., 2011). Given this, assessing the extent of both intentional and incidental representation of freshwaters within existing protected area networks is a major prerequisite for identifying and plugging conservation gaps (Herbert et al., 2010).

To date, most conservation efforts have focused on protecting areas that ensure adequate representation of taxonomic diversity (TD), such as species richness (e.g. Rodrigues et al., 2004).

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Despite this, it is increasingly well recognized that protected areas should strive to preserve all components of biodiversity including the ecological and evolutionary processes that generate and maintain it and the goods and services that humans obtain from nature (Mulongoy and Chape, 2004). Functional diversity (FD), which reflects the range of biological, physiological and ecological traits within natural communities (Petchey and Gaston, 2006), has been advocated as an important facet of diversity for ensuring the provision of goods and services (Díaz et al., 2006), and has been shown to be a key driver of ecosystem processes (e.g. Mokany et al., 2008); essential in understanding relationships between biodiversity, ecosystem functioning and environmental constraints (Mouchet et al., 2010). In addition, incorporating functional information into conservation strategies allows for this approach to go beyond simple species representation. Indeed, human activities may have an impact on FD and alter species interactions and ecosystem functioning regardless of the change in taxonomic diversity (Díaz et al., 2006). Similarly, regions of high TD may be incongruent with regions of high FD (Cumming and Child, 2009), and such spatial mismatch between different aspects of diversity may result in protected area networks that do not fully represent biodiversity (Abellán et al., 2013; Devictor et al., 2010).

Effective protected area systems in a changing world should also ensure the maintenance of species and functional processes at different spatial scales (Brooks et al., 2006; Devictor et al., 2010; Gering et al., 2003). Both taxonomic and functional diversity can be broken down into local, regional and among-site components (so-called alpha, gamma and beta diversities; Whittaker, 1972). Whilst the effectiveness of protected areas in representing alpha (e.g. species richness in a given site, or local diversity) and gamma biodiversity (the total taxa represented in a protected area network, or regional diversity) have often been assessed (e.g. Araújo et al., 2007; Brancquart et al., 2008; Rodrigues et al., 2004), very few studies have explored how well existing protected area networks represent beta diversity (and its components), despite the fact that it is the rate of species (or trait) turnover between sites that dictates the optimal spatial arrangement of conservation areas (Nekola and White, 2002). As beta diversity quantifies the change in species (or traits) across space, it provides information about variation in species assemblages, which can be very useful to preserve ecological and evolutionary processes as well as the underlying environmental heterogeneity necessary for long-term persistence (Fairbanks et al., 2001; Margules and Pressey, 2000). Furthermore, beta diversity itself is comprised of two components: spatial turnover and nestedness (see Baselga, 2010). Whilst both nestedness (i.e. a pattern characterised by depauperate sites being strict subsets of richer ones) and turnover (i.e. species/trait replacement from site to site) are components of beta diversity, they have different conservation implications (Wright and Reeves, 1992). A preponderance of nestedness within a network would permit the prioritization of just a small number of the richest sites, whilst high turnover would require conservation of a larger number of different sites, not necessarily the richest ones (Baselga, 2010).

In this study we investigate how effectively protected area networks, not specifically designated for freshwaters, support aquatic macroinvertebrate biodiversity in the Iberian Peninsula. We explore the behaviour of both taxonomic and functional diversity measures, at local (α), inter-site (β) and regional (γ) scales. In addition to making up the bulk of freshwater animal biodiversity, macroinvertebrates play a key role in freshwater ecosystem processes (Covich et al., 1999) but are still less studied and protected compared to other, more publicly appealing, taxa (Strayer, 2006). We specifically use water beetles as a surrogate of overall macroinvertebrate diversity, as these represent one of the most diverse and best known groups of aquatic invertebrates in the region (Millán

et al., 2014; Ribera, 2000), living across the complete spectrum of inland water habitats. Aquatic beetles have been shown to be excellent surrogates for wider macroinvertebrate biodiversity in Iberia (Guareschi et al., 2012; Sánchez-Fernández et al., 2006) and elsewhere (Bilton et al., 2006) and have been used to select priority areas for aquatic conservation (Foster et al., 1989; Sánchez-Fernández et al., 2004). Furthermore, aquatic Coleoptera show considerable diversity in life histories and ecological strategies (Jäch and Balke, 2008), and are therefore functionally important in aquatic habitats, being involved in ecosystem processes such as biomass production, nutrient cycling and recourse processing.

Specifically, we use aquatic beetle data from the Iberian Peninsula to address the following questions: (i) do protected areas have significantly higher α -diversity than non-protected areas? (ii) do protected area networks include more total diversity (γ) than expected by chance, given their area? and (iii) do protected area networks include more inter-site diversity (β) than expected by chance alone, given their area? Addressing the first question we explore whether protected areas include those cells with the highest diversity (e.g. species richness), whilst answering the second question provides information about the effectiveness of the whole network in representing overall freshwater diversity. Finally, the third question deals with dissimilarity amongst protected sites within the network, which is related to their complementarity, a principle widely used in conservation planning (Justus and Sarkar, 2002). Our study has wider implications for the design of protected area networks, being the first investigation to explore how well such networks support both taxonomic and functional measures of biodiversity in a non-target group across different spatial scales.

2. Methods

2.1. Study area and data

This study focuses on the Iberian Peninsula, a biodiversity hotspot located in south-western Europe, which is mainly composed of the mainland territories of Portugal and Spain (Fig. 1). The region, which extends nearly 600,000 km², includes a variety of biomes, relief, climates, and soil types, where altitude ranges from sea level to 3483 m. The study area is one of the richest European regions in terms of animal species diversity (Williams et al., 2000) and is characterised by a wide variety of ecosystem types, including aquatic environments, some of which are rare on a European context (Millán et al., 2011).

Distributional data of Iberian water beetles at 10 × 10 km resolution were obtained from the ESACIB database (see Sánchez-Fernández et al., 2008a; Millán et al., 2014), which represents the most complete information available for a group of freshwater macroinvertebrates in the study area. The database currently contains over 60,000 records with associated location data (10 × 10 km UTM cells) for 484 water beetle species. Species level was used for taxonomic diversity measures whilst genus level information was used to assess functional diversity (Dolédéc et al., 2000; Gayraud et al., 2003).

Two different protected area networks were investigated: the extant regional and national protected area network (RNAs) and the wider and incompletely implemented Natura 2000 network of protected areas (N2000) (see Fig. 1). RNAs are at the core of national and regional conservation policies, and include National and Natural Parks, Natural Reserves, Natural Monuments, Protected Landscapes, Protected Marine Areas (not included in this study) as well as different types of local protected areas (i.e. those included in Spanish and Portuguese laws). The N2000 network

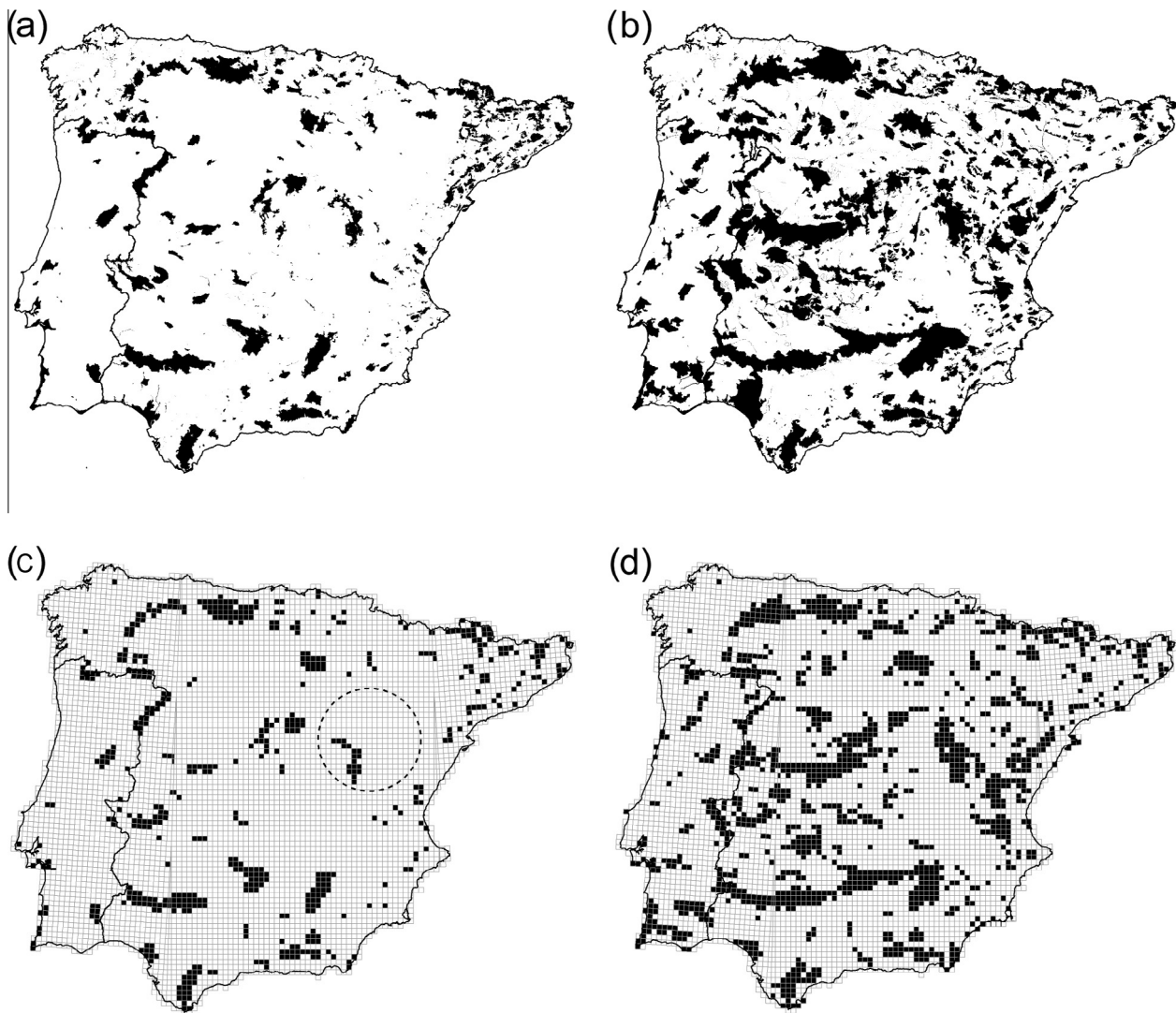


Fig. 1. Study area (Iberian Peninsula) showing the two different protected area networks considered (a) regional and national protected areas (RNAs), (b) Natura 2000 (N2000) and the cells considered as protected by RNAs (c) or N2000 (d) at a threshold of 50% coverage. Dashed circle in (c) represents an example of a 100 km radius window.

represents a key tool for biodiversity conservation at the European scale and aims to guarantee the long-term survival of Europe's most valuable and threatened species and habitats. It includes Special Areas of Conservation (SACs) designated under the Habitats Directive (92/43/EC), and Special Protection Areas (SPAs) which are designated under the Birds Directive (79/409/EC, 2009/147/EC). GIS data layers supplied by the national conservation agencies of Spain and Portugal, were edited and combined to produce single layers of Iberian RNAs and N2000 networks, respectively.

Since species distribution information is available at a resolution 10×10 km UTM cells, whereas protected areas are polygons, often smaller than these grid cells, we applied a filter to identify grid cells that can be considered protected (see Araújo et al., 2007; Sánchez-Fernández et al., 2013 for similar approaches). We calculated the area of each 10×10 km UTM cell included within the two different protected area scenarios and a cell was considered protected when at least 50% of its area was within RNA or N2000 sites (Fig. 1). This threshold can be considered appropriate, as most aquatic habitats are highly influenced by processes occurring in their catchments and similar thresholds have been used elsewhere in comparable evaluations (e.g. Araújo, 2004; D'Amen

et al., 2013; Sánchez-Fernández et al., 2013). Nevertheless, to assess the sensitivity of results to this choice of threshold alternative values (1%, 25%, 75% and 90%) were also considered (see Appendix A in Supplementary materials).

2.2. Diversity measures

We calculated values describing the different components (α , β and γ) of water beetle TD and FD included in protected areas. Thus, for each protected area network (RNAs or N2000) we assessed the diversity (either TD or FD) within each protected cell (α diversity), the change in diversity amongst protected cells (β diversity) and the total diversity in the whole network of protected cells (γ diversity). Both α and γ TD were calculated as species richness, whilst β TD was measured as the change in species composition amongst sites (grid cells in our case). β TD was measured as both spatial turnover in species composition (β_{SIM} Simpson's dissimilarity) and variation in species composition due to nestedness (β_{NES} nestedness-driven dissimilarity), with overall β diversity (β_{SOR} , Sørensen's dissimilarity) being the sum of these components (Baselga, 2010). For a given set of grid cells, overall β diversity

and its two components were calculated using multiple-site dissimilarity measures following Baselga (2010).

To characterise the functional diversity of coleopteran communities, 11 biological traits (considering 40 modalities, see Appendix A) were considered following Tachet et al. (2010), which relate to morphology, life history, dispersal ability and trophic role. Tachet et al.'s trait database summarises expert knowledge on European freshwater invertebrates by assigning an affinity score for each taxon to each modality using a fuzzy coded approach. Selection of values was conducted using information developed for Mediterranean ecosystems by Bonada and Dolédec (2011) and Picazo et al. (2012), with some scores being modified on the basis of the authors' own fieldwork.

A matrix of trait pairwise dissimilarity among taxa was generated using Gower distance, and a functional dendrogram constructed from this distance matrix using Ward linkage and hierarchical clustering (Petchey and Gaston, 2002 – alternative methods gave qualitatively similar results). Subsequently FD was calculated for each grid cell (α diversity) and for each protected area network (γ diversity), as the mean pairwise distance amongst species – i.e. the average of the branch lengths in the trait dendrogram joining all pairs of species in a given community (Webb et al., 2002). Furthermore, functional β diversity was computed using the Functional Sørensen's Index (F_{SOR} ; Swenson et al., 2011), which is an analogue of the traditional Sørensen's Index. F_{SOR} is a dissimilarity metric that provides an overall indicator of the shared function between two communities and, like its phylogenetic analogous metric (PhyloSor, Bryant et al., 2008), can be broken down into both turnover (F_{sim}) and nestedness (F_{nes}) components of functional β diversity (Leprieur et al., 2012). We therefore also partitioned these components of β diversity using the framework proposed by Baselga (2010).

2.3. Assessment of protected areas performance

In order to assess the performance of protected area networks in representing γ and β taxonomic and functional freshwater diversity, we compared the level of TD and FD representation within both networks with what would be expected by chance alone, given the area they cover. To do this, we compared observed γ and β TD or FD to expected values from 1000 random draws of an equal number of cells, taken from the pool of grid cells where species have been recorded (including both protected and unprotected cells). The proportion of random samples with higher or lower diversity values than those observed for protected area networks allowed us to obtain a significance value (see e.g. Araújo et al., 2007). In the case of α TD and FD, we assessed whether values of TD and FD in protected grid cells were significantly greater than in unprotected cells using nonparametric Wilcoxon–Mann–Whitney tests.

We investigated the effectiveness of protected area networks in representing macroinvertebrate biodiversity at two different spatial scales. First, we assessed α -, β - and γ -diversity represented by the whole reserve networks for the entire Iberian Peninsula. Second, as network planning is often carried out at an intermediate spatial scale, we assessed protected area performance at a sub-regional scale according to the following procedure (see also Devictor et al., 2010): a circle with a radius of 100 km (Fig. 1) was centred on each 10 × 10 km cell (so that all grid cells are the centre of one 100 km radius window – such windows are overlapping and therefore some plots included in one window will be included in neighbouring ones). Then, for each of these 100 km radius windows we calculated α -, β - and γ -diversity in protected cells and compared them with random expectations as described above (for example, we compared the observed γ -diversity of protected cells within a given window to the values expected from

1000 random draws of an equal number of grid cells within that window).

All analyses were performed using R (R-Development-Core-Team, 2010) with packages “ade4” (Chessel, 2011), “Betapart” (Baselga and Orme, 2012), “picante” (Kembel et al., 2010) and “FD” (Laliberté and Legendre, 2010).

3. Results

3.1. Representation of taxonomic diversity in protected areas

In terms of α diversity at the Iberian Peninsula scale, species richness of water beetles was significantly higher in protected than in unprotected cells for both RNAs and N2000 (Wilcoxon–Mann–Whitney test $P < 0.001$ – see Appendix A). Furthermore, Iberian protected areas (both RNAs and N2000) represented significantly more total species (γ diversity) than expected given the area they cover. In total, RNA cells ($n = 352$) included 419 water beetle species (86.6% of the Iberian species pool) and this value was significantly higher than those obtained by chance (Table 1). This pattern is even clearer for N2000 cells ($n = 690$) that supported 458 species (94.6% of the Iberian species pool).

Overall β TD of water beetles in RNAs ($\beta_{\text{SOR}} = 0.9940$) mainly results from spatial turnover ($\beta_{\text{SIM}} = 0.9866$), with only a small contribution from nestedness (Table 2). Despite the high value of overall β diversity, it was significantly lower than values generated by random sampling of cells ($\beta_{\text{SOR}} = 0.9944 \pm 0.0001$ in random samples; $P = 0.002$). In terms of the two components of β diversity, observed nestedness was significantly lower than random values, whilst turnover was not significantly different from random expectations (Table 2). A similar pattern was found for N2000, with overall β diversity mainly due to turnover. Again, both β_{SOR} and β_{NES} were significantly lower than expected by chance, whilst in this case β_{SIM} was significantly higher than random (Table 2).

At a smaller spatial scale, i.e. in the 100 km radius windows, α -diversity was significantly higher in protected cells than in unprotected ones for 50% of windows in the case of RNAs, and 68% in the case of the N2000 network. Windows with non-significant differences in α TD were mainly located in Eastern Iberia and along the central boundaries between Portugal and Spain (Fig. 2a and d). When we focus on γ diversity within windows, an opposite and more marked pattern was detected: for most of the windows (86% for RNAs; 83% for N2000) protected area networks did not perform significantly better in term of total species richness than an equal number of random cells within the same windows (see also Fig. 2b and e).

Focussing on β diversity, again for most of the windows (99.6%) overall β diversity among RNAs protected cells (average $\beta_{\text{SOR}} = 0.9082 \pm 0.0528$ across windows) was mainly caused by spatial turnover ($\beta_{\text{SIM}} = 0.7939 \pm 0.1051$), with only a small contribution from nestedness ($\beta_{\text{NES}} = 0.1143 \pm 0.0698$). Furthermore,

Table 1

Representation of taxonomic and functional γ diversity in protected area networks at Iberian scale (protected) and comparison with the values expected from 1000 random draws of an equal number of grid cells (random, mean value $\pm$ SD). Standardized effect sizes (SES; which describe the difference between the observed value of diversity in the protected area network versus the mean value obtained from random draws, divided by the standard deviation of null data) and the p values are shown.

	Protected	Random	SES	p
<i>Taxonomic diversity</i>				
RNAs	419	404.2 $\pm$ 8.4	1.76	0.04
N2000	458	442.7 $\pm$ 5.9	2.60	0.002
<i>Functional diversity</i>				
RNAs	0.283	0.283 $\pm$ 0.002	0.00	0.692
N2000	0.284	0.283 $\pm$ 0.001	1.00	0.131

Table 2

Representation of taxonomic and functional beta diversity (β_{SOR}) and its decomposition in nestedness (β_{NES}) and turnover (β_{SIM}) in protected area networks at Iberian scale (protected) and comparison with the values expected from 1000 random draws of an equal number of grid cells (random). Standardized effect sizes (SES; which describe the difference between the observed value of diversity in the protected area network versus the mean value obtained from random draws, divided by the standard deviation of null data) and the p values are shown.

	RNAs				N2000			
	Protected	Random	SES	p	Protected	Random	SES	p
<i>Taxonomic diversity</i>								
β_{SOR}	0.9940	0.9944	−4.20	0.002	0.9969	0.9970	−9.61	0.002
β_{SIM}	0.9866	0.9859	1.34	0.180	0.9928	0.9925	1.80	0.038
β_{NES}	0.0074	0.0086	−0.28	0.022	0.0040	0.0046	−3.02	0.008
<i>Functional diversity</i>								
β_{SOR}	0.9890	0.9907	−11.43	0.002	0.9946	0.9952	−12.56	0.002
β_{SIM}	0.9671	0.9722	−4.76	0.002	0.9835	0.9855	−5.63	0.002
β_{NES}	0.0220	0.0186	3.43	0.001	0.0111	0.0097	4.20	0.002

RNA protected cells did not perform especially well in terms of overall β TD; 61.3% of windows having values of β_{SOR} in protected cells significantly lower than random expectations, and just 0.04% of windows having significantly higher values for protected cells (Fig. 2c). In the case of turnover, only 2.1% of windows had significantly higher β_{SIM} values in protected cells than random expectations (these cells being clustered between Galicia and the Cantabrian Mountains, at north and north-western of Iberian Peninsula), whilst 19.6% had significantly lower values (Fig. 3a). Finally, 4.3% of the windows had significantly higher nestedness in their network of protected cells than expected by chance, whilst 4.0% had significantly lower values (again grouped between Galicia and the Cantabrian Mountains; Fig. 3b).

In the case of N2000, similar patterns of β diversity were obtained. Overall β diversity amongst N2000 protected cells ($\beta_{\text{SOR}} = 0.9445 \pm 0.0276$) was mainly (99.9% of windows) due to spatial turnover ($\beta_{\text{SIM}} = 0.8699 \pm 0.0673$ versus $\beta_{\text{NES}} = 0.0746 \pm 0.044$). Here 66.2% of windows had β_{SOR} values significantly lower in the network of protected areas than random expectations, and none had significantly higher values (Fig. 2f). 14.8% of windows had significantly lower β_{SIM} values in the protected network than expected by chance (clustered in the Cantabrian Mountains), and just 0.09% higher (Fig. 3e). Finally, 3.6% of windows had significantly higher values of β_{NES} in their network of protected cells, whilst 0.6% had significantly lower values than those obtained by chance (Fig. 3f).

All these results were basically consistent across thresholds used to consider a cell as protected (see Appendix A).

3.2. Representation of functional diversity in protected areas

In terms of α diversity at the Iberian Peninsula scale, FD was significantly higher in protected (both RNA and N2000 networks) than in unprotected cells (Wilcoxon–Mann–Whitney test; $P < 0.001$; Appendix A). However, both protected area networks did not include significantly more γ FD than expected by chance at this scale (Table 1). Again, in the case of β diversity, for both RNAs and N2000 overall β diversity mainly resulted from spatial turnover (β_{SIM}), with only a small contribution from nestedness. For both protected area networks, β_{SOR} and β_{SIM} were significantly lower than random expectations, whilst β_{NES} was significantly higher (Table 2).

At a smaller spatial scale, i.e. in the 100 km windows, α FD was significantly higher in protected cells than in unprotected ones for just 26% of windows in the case of RNAs, and 30% in the N2000 network. These windows were basically located in southern Iberia, northern Iberian (Cantabrian Mountains) and the Pyrenees (Fig. 4a and d). Furthermore, in most of the windows (98.6% for RNAs and 95.3% for N2000) protected area networks did not

perform significantly better in term of γ FD than expected by chance. In the case of N2000, windows with significantly higher values of γ FD were mainly clustered in the southernmost part of the Iberian Peninsula (Fig. 4e). Focussing on β FD, again for the most of the windows (88%), overall β diversity amongst RNA protected cells ($\beta_{\text{SOR}} = 0.8019 \pm 0.1015$) was mainly due to spatial turnover ($\beta_{\text{SIM}} = 0.5542 \pm 0.1648$), with a smaller contribution from nestedness ($\beta_{\text{NES}} = 0.2477 \pm 0.0981$). 50.9% of windows had values of β_{SOR} significantly lower in RNA protected cells than random expectations, and no windows had significantly higher values (Fig. 4c). Also, no windows had significantly higher β_{SIM} values amongst protected cells than random, whilst 30.4% had significantly lower values (mainly clustered in the Cantabrian Mountains and the south of Spain, Fig. 3c). Finally, 15.0% of windows had significantly higher values of β_{NES} amongst protected cells (with a similar geographic pattern, Fig. 3d), whilst 1.17% had significantly lower values than random expectations.

With the Natura 2000 network, similar patterns of β FD were obtained: for most of the windows (98.7%) overall β diversity (average $\beta_{\text{SOR}} = 0.8858 \pm 0.0674$ across windows) was mainly driven by spatial turnover (average $\beta_{\text{SIM}} = 0.7135 \pm 0.1156$ versus $\beta_{\text{NES}} = 0.1724 \pm 0.0638$). 56.9% of windows had values of β_{SOR} significantly lower in the N2000 network, and no windows had significantly higher values for protected cells (Fig. 4f). Only 0.34% of windows had significantly higher β_{SIM} values in protected cells than random, whilst 36.8% had significantly lower values (Fig. 3g). Finally, 24.2% of windows had significantly higher values of β_{NES} in their network of protected cells, whilst 1.3% had significantly lower values than expected by chance (Fig. 3h).

Again, all these results were consistent across the different thresholds used to consider a cell as protected (see Appendix A).

4. Discussion

Our research represents one of the first attempts to assess the effectiveness of protected areas in representing different components of invertebrate biodiversity at different spatial scales, despite the fact that invertebrates make up the bulk of global animal diversity, in both taxonomic and functional senses. Ramos et al. (2001) have estimated that about 98% of the total Iberian fauna is made up of invertebrates, for example, and similar values apply elsewhere. Biodiversity knows no political boundaries and in this context evaluations of protected areas at biogeographical scales (like the Iberian Peninsula) are desirable, but as we demonstrate these should be combined with studies at different scales which will assist in the design and management of protected areas.

Our results show that, at the Iberian scale, protected areas (both RNAs and N2000) tend to include those cells of highest diversity (both taxonomic and functional), as revealed by α diversity

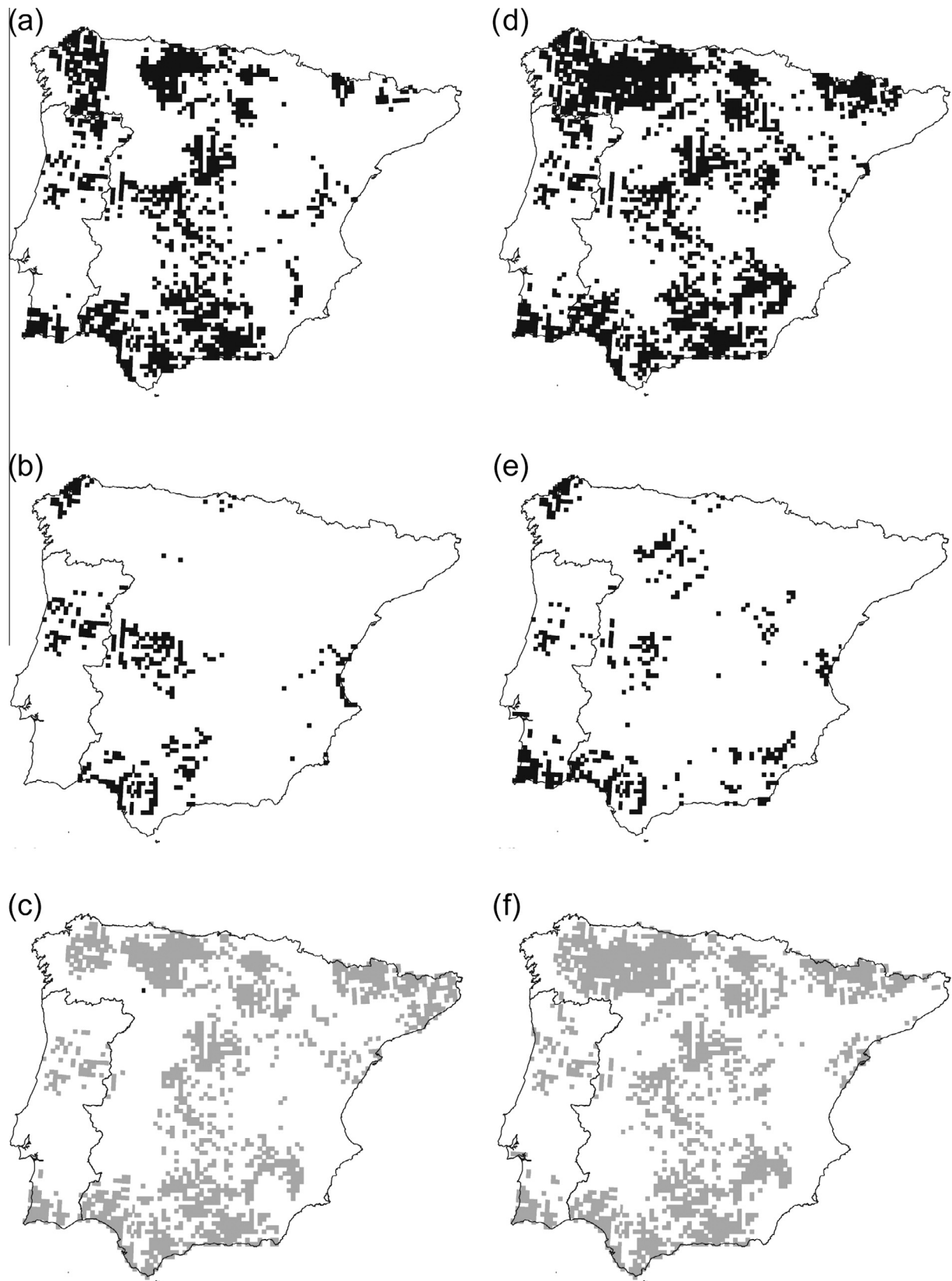


Fig. 2. Performance of RNAs (a–c) and N2000 (d–f) networks in representing α , β and γ taxonomic diversity at subregional scale (100 km radius windows): (a and d) windows for which α diversity is significantly higher in protected than in unprotected cells; (b and e) windows for which γ diversity was significantly higher than random; (c and f) windows for which overall beta diversity (β_{SOR}) was significantly higher (black cells) and lower (grey cells) than random. Each grid cell represents the centre of the window.

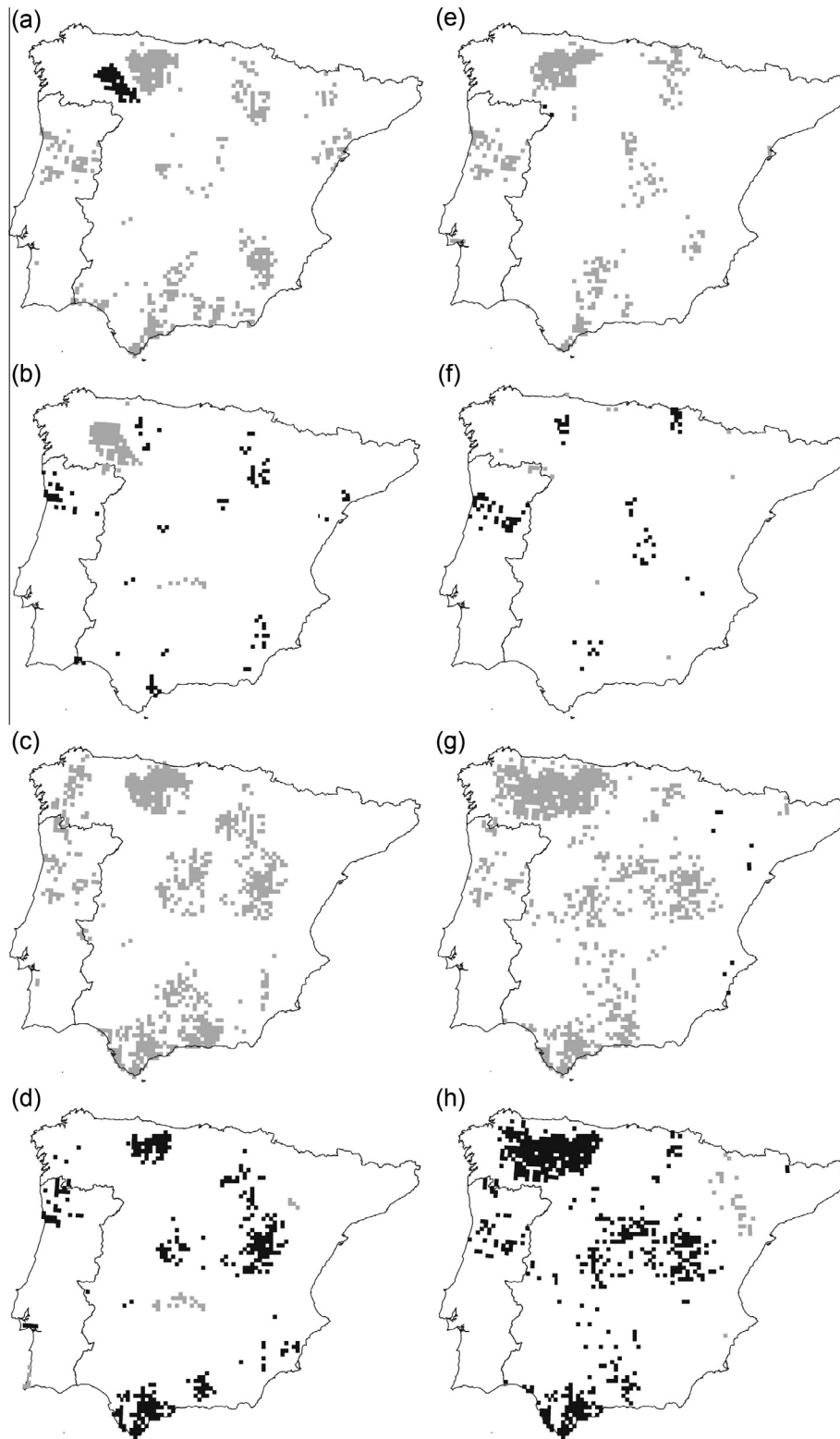


Fig. 3. Performance of RNAs (a–d) and N2000 (e–h) networks in representing taxonomic and functional beta diversity components (turnover and nestedness) at subregional scale (100 km radius windows). Grid cells representing the centre of those 100-km-radius windows with significantly higher (black cells) and lower (grey cells) values than random are shown. (a and e) taxonomic β_{SIM} ; (b and f) taxonomic β_{NES} ; (c and g) functional β_{SIM} ; (d and h) functional β_{NES} .

assessments. Protected areas usually tend to over-represent mountainous areas and under-represent lowlands, probably because the former provide less conflict with human land uses (Joppa and Pfaff, 2009). This is true for the two studied protected area networks in the Iberia Peninsula (mean altitude of RNA cells 892 m versus

588 m in unprotected ones, and 807 m in N2000 cells versus 561 m in unprotected ones; Wilcoxon–Mann–Whitney test, $P < 0.001$ in both cases), something which is common in many countries (e.g. Maiorano et al., 2007). These mountainous areas usually include a high density of relatively unimpacted localities

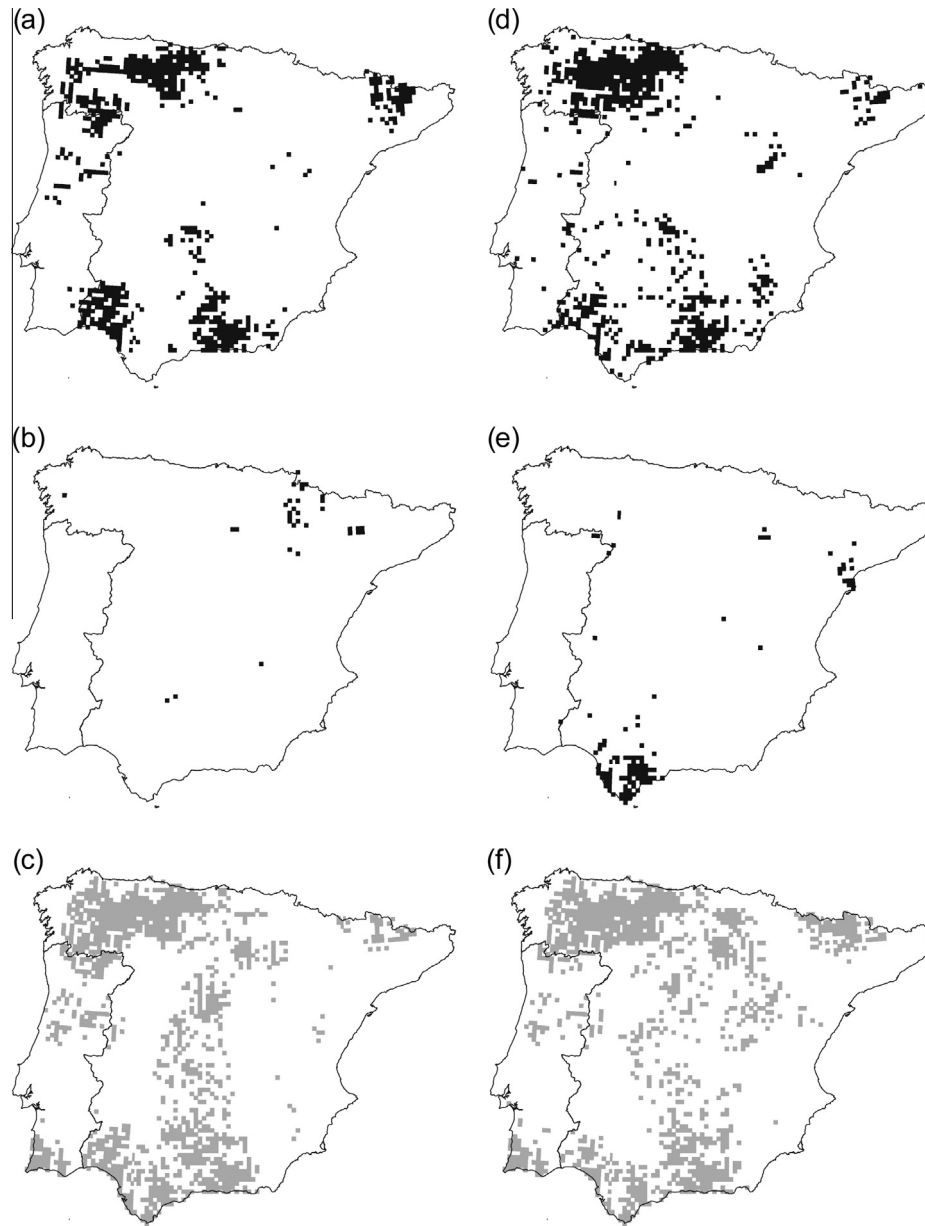


Fig. 4. Performance of RNAs (a–c) and N2000 (d–f) networks in representing α , β and γ functional diversity at subregional scale (100-km-radius windows). (a and d) windows for which α diversity is significantly higher in protected than in unprotected cells; (b and e) windows for which γ diversity was significantly higher than random; (c and f) windows for which overall beta diversity (β_{SOR}) was significantly higher (black cells) and lower (grey cells) than random. Each grid cell represents the centre of the window.

with, in general, the highest water beetle diversity (Ribera, 2000; Millán et al., 2014). Nevertheless, it should be noted here that a high local diversity in a given site does not necessarily indicate high conservation value. For instance, some habitats support few species, but contain many endemic or exclusive taxa. Similarly, local diversity is often maximized at a certain level of human disturbance (Wilkinson, 1999), rather than in pristine sites.

When we focus on γ diversity, protected area networks capture a relatively large proportion of the Iberian water beetle species pool (87% and 95% in RNAs and N2000, respectively); significantly more than expected by chance, given their area, despite the fact that they were established with little or no consideration of their freshwater biodiversity. These results are congruent with those for Iberian plant and terrestrial vertebrate species (Araújo et al., 2007), which show that, in general, protected areas conserve more species than expected by chance. Nevertheless, it should be noted

that the distribution and extent of existing reserves is still inadequate or insufficient to protect some water beetles of conservation concern in the study area (see also Sánchez-Fernández et al., 2008b). Among the 64 species (13%) that occur completely outside the RNA network, 14 are Iberian endemics with high vulnerability (Millán et al., 2014; Sánchez-Fernández et al., 2008b). Moreover, in the case of the 25 species excluded from the N2000 network (5%), four (*Iberoporus cermenius* Castro & Delgado, 2001, *Nebrioporus croceus* Angus, Fresneda & Fery, 1992, *Rhithrodytes agnus argaensis* Bilton & Fery, 1996 and *Ochthebius irenae* Ribera & Millán, 1999) are also vulnerable Iberian endemics. In a similar vein, a recent study has shown that current protected sites in Spain do not allow a complete representation of endangered invertebrates (Hernández-Manrique et al., 2012). Most of the detected gap-species (i.e. those not represented in any protected area at a threshold of 50%) are distributed along the Mediterranean coast

and eastern Iberia, inhabiting small lentic, coastal or/and mineralized ecosystems (Fig. A1) and their localities should be considered for future protection. Interestingly, these patterns were not congruent for α and γ diversity at smaller scale, with a considerable percentage of 100-km-radius windows (50–30% for TD and ca. 70% for FD) showing no significant differences in α diversity between protected and unprotected areas, and most (80–95%) with no significant differences in total species representation (γ diversity). This suggests that in some areas of the Peninsula protected areas are not biased towards the most speciose localities or zones, and they do not cover sufficiently subregional diversity. The size of our 100-km-radius windows can be considered as a valid approximation to the scale of conservation management that often occurs on the ground, being more realistic in many cases than a biogeographical or national scale. For example, in many countries with federal distribution of powers, each region may have its own system of protected areas (EEA, 2012). In the case of Spain, most nature conservation responsibilities are devolved to regional autonomous governments (Morillo and Gómez-Campo, 2000), and most of current protected areas have been designated under this regional setting. The mean size of the Spanish autonomous regions is around 30,000 km², approximating to that of our windows (31,400 km²).

In contrast to TD, the overall FD included in both protected area networks was not significantly higher than chance expectations, given the areas they cover. In other words, the level of FD representation achieved is no better than that expected by chance. Furthermore, this pattern was consistent across the two spatial scales for γ diversity. These findings illustrate the relative failure of both protected area systems in preserving freshwater functional diversity, despite this component of biodiversity being a better predictor of ecosystem productivity and vulnerability than species diversity (Hooper et al., 2005) and illustrate the importance of explicitly taking all types of diversity into account when designing protected areas (Cumming and Child, 2009; Devictor et al., 2010; Forest et al., 2007; Strecker et al., 2011).

This incongruence in the performance of protected areas in representing either TD or FD could result from a bias in habitat representation within reserve networks. This could result, for example, if certain aquatic habitats (e.g. headwaters) are overrepresented within protected areas, and these contain relatively higher species diversity than functional diversity (the correlation between TD and FD across Iberian protected cells was low; Pearson's correlation coefficient $r = 0.29$ and $r = 0.32$ for RNAs and N2000, respectively; $P < 0.001$ in both cases). This appears to be the case of headwater streams in mountain areas, which have a rich, and often locally endemic, but functionally specialized, and rather homogenous fauna (Picazo et al., 2012). This situation could be due to environmental and biogeographical factors affecting the two diversity components in different ways. Environmental filtering will limit community members to those that are preadapted, and thus functionally similar, creating a functional structure of communities that is not necessarily reflected in their taxonomic structure (Poff, 1997). Our results suggest that important ecosystem processes (e.g. energy flow, nutrient cycling and biomass production) and services (e.g. mosquito control, food source to species of commercial interest) in which water beetles (and aquatic macroinvertebrates in general) are involved in some lowland freshwater habitats could remain under-represented by protected areas.

In case of β diversity (both TD and FD) at the Iberian scale, although the overall value obtained in both protected area networks was very high (close to 1), it was always significantly lower than expected by chance – i.e. communities in protected areas were more similar than would be expected from a random draw of equivalent area. This result may reflect the wide environmental heterogeneity in the Peninsula, which, despite being reasonably

well represented in protected areas, may still be insufficient to cover the full spectrum of environments and associated faunas. Deconstructing β diversity into its spatial turnover and nestedness components provides crucial insights here, showing that spatial turnover is responsible for most β diversity amongst protected sites, with only a small contribution from nestedness. β diversity within Iberian reserve systems is therefore mainly the result of species/trait replacement between sites. Although rarely recognized, the principles behind most approaches to systematic planning, such as complementarity, are driven by patterns of β diversity (Magurran, 2004; Justus and Sarkar, 2002). As the turnover component of β diversity (β_{SIM}) quantifies the change in biodiversity attributes across space, it provides critical information to represent all biodiversity within practical constraints such as area and cost (Margules and Pressey, 2000). Ideally, protected area networks should maximize the species/trait turnover amongst protected sites within the network, as this would denote high complementarity among protected sites. On the other hand, the nestedness component of β diversity (β_{NES}) provides information about redundancy in biodiversity attributes in the network, which also may be relevant for the persistence of biodiversity over time, as multiple representations of species within the set of sites would increase the probability of regional persistence (i.e. maintain each species within the reserve network, even if local extinctions occur (Cabeza and Moilanen, 2001).

When we compared both spatial turnover and nestedness components with expectations based on the surface covered by protected areas, we again found contrasting results between TD and FD. Whilst protected area networks showed significantly less turnover and more nestedness than random for FD, they generally displayed the opposite pattern for TD, suggesting that the processes underlying these β diversity values are different. Spatial turnover implies the replacement of some biodiversity attributes by others as a consequence of environmental sorting or spatial and historical constraints (Qian et al., 2005). The fact that we found lower functional turnover amongst protected areas than expected at random, despite higher taxonomic replacement, seems to suggest that spatial turnover within reserve networks is mainly driven by spatial and historical constraints and less so by environmental sorting. The same would apply for species/traits loss (or gain) between protected sites, which suggest a higher functional redundancy within reserve systems than expected at random, despite lower taxonomic redundancy. These results point again to a bias in environmental representation within protected area networks, in agreement with γ and α diversity assessment. Although the results for 100-km-radius windows were more equivocal, they generally point to the same pattern. They also show that the performance of protected area networks in representing β diversity at this smaller spatial scale differs between regions of the Iberian Peninsula, since windows significantly departing from random were usually geographically clustered.

Finally, it is worth noting that our results did not show important qualitative differences between the two studied protected area networks. The complete implementation of the proposed Natura 2000 network may mean an important increase of the area of protected land in the study area (e.g. from around 12% of RNAs up to 28% in Spain; Europarc-España, 2010), with a subsequent increase in the taxonomic diversity represented at Iberian scale (e.g. up to 95% of the Iberian species pool). However, and despite the important potential of this reserve system enhancing the protection of European endangered species and habitats (European Commission, 1992) we found that Natura 2000 network did not often over performed RNAs in relative terms, this is, when compared to that would be expected given the area it covers (for instance, in the conservation of γ functional diversity). These findings highlight some limitations of this network in preserving

numerous components of freshwater biodiversity (see also Hermoso et al., 2014; Sánchez-Fernández et al., 2013) and the need of explicitly consider their biota and environments into conservation planning at European level (Hermoso et al., 2014). In any case, it is worth noting also that, in the case of inland waters, the inclusion in protected areas does not automatically guarantee protection because they often fail to address important aquatic concerns such as whole-catchment integrity, hydrology, and introductions of non-native species (Moyle and Randall, 1998). In this context the expansion of protected areas beyond their terrestrial comfort zone and the change towards catchment-based conservation planning (e.g. Nel et al., 2009; Bruno et al., 2014) seems crucial if we aim to improve protected area performances in terms of freshwater biodiversity conservation.

5. Conclusions

Our approach represents a synthetic view of the performance of protected area networks in representing different facets (taxonomic and functional) and components (α , β and γ) of water beetles biodiversity at different spatial scales.

Overall, our findings highlight the contrasting performance of reserve systems in the maintenance of either taxonomic and functional diversity, as well as the importance of spatial scale. First, results show that, despite the fact that Iberian reserves were often set up without any consideration of non-target organisms, they perform relatively well for taxonomic diversity of water beetles at Iberian Peninsula scale. Although the extent to which these results are applicable to other geographical areas and freshwater taxa remains to be investigated, they seem to highlight the potential role of protected area systems in representing the taxonomic diversity of non-target groups.

Second, our results show a poor performance of Iberian protected areas in representing the functional diversity of water beetles at Iberian scale, in contrast to findings for taxonomic diversity. Such mismatch between results for taxonomic and functional diversity call for caution against the use of any one diversity component as a surrogate for others, and emphasize the importance of adopting an integrative approach to biodiversity conservation in aquatic ecosystems (Abellán et al., 2013; Devictor et al., 2010; Strecker et al., 2011). And third, our results point contrasting results at smaller spatial scale, highlighting the importance to consider the influence of spatial scale when evaluating the effectiveness of protected area systems (Andelman and Willig, 2002; Caro et al., 2009; Gaston et al., 2006).

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Appendix A. Supplementary material

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.biocon.2015.04.018>.

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